

Reversed and Non-Equivalent Ratios

Answer Key

Grades 6–7

Quick Answers

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|------------------|------|----------|
| 1. $\frac{3}{2}$ | 4. 8 | 7. 8 |
| 2. — | 5. 9 | 8. —, 20 |
| 3. $\frac{3}{2}$ | 6. — | |
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Curriculum

Level: Grades 6–7

6.RP.A.2, 6.RP.A.3 – *problems 1, 2, 3, 4, 5, 6, 7, 8*

Difficulty 1–4 of 5 across 12 tagged checks.

Common wrong answers

5. *If they got 8: added 2 to each part instead of multiplying both parts by the same factor, giving 8 scoops*

7. *If they got 0.125: divided the minutes by the pages instead of the pages by the minutes, reporting 0.125 pages per minute*

1. Flour to sugar. The question asks for flour first, so flour goes on top: 3 cups of flour to 2 cups of sugar is $\frac{3}{2}$. It is already in simplest form, since 3 and 2 share no common factor. Sugar to flour would be $\frac{2}{3}$ — a different comparison, not a different way of writing the same one.

2. Are 4 : 6 and 6 : 9 equivalent? Simplify each by dividing both parts by their greatest common factor: $\frac{4}{6} = \frac{2}{3}$ (divide by 2) and $\frac{6}{9} = \frac{2}{3}$ (divide by 3). Both reduce to the same ratio, so $\boxed{=}$: they are equivalent. Cross products confirm it: $4 \times 9 = 36$ and $6 \times 6 = 36$.

3. Find the mistake (Leo). (a) A ratio names its quantities in the order asked. Writing $\frac{12}{18}$ puts boys on top, so it answers “boys to girls”, a different comparison from the one requested. [Manual review: any sentence saying the first quantity named goes on top.]

(b) Girls to boys means girls on top: $\frac{18}{12}$. Divide both parts by 6: $\frac{3}{2}$. Sense check: there are more girls than boys, so a correct girls-to-boys ratio must be greater than 1. Leo’s $\frac{2}{3}$ is less than 1, which is the quickest way to catch a reversed ratio.

4. Pencil table. The two columns must be equivalent ratios, so

$$\frac{3 \text{ pencils}}{2 \text{ dollars}} = \frac{12 \text{ pencils}}{x \text{ dollars}}$$

Cross multiply: $3x = 2 \times 12 = 24$, so $x = 8$ dollars. Scale-factor check: 3 pencils became 12 pencils, a factor of 4, so the dollars must also be multiplied by 4: $2 \times 4 = 8$. Both parts scale by the SAME factor, which is what makes the columns equivalent.

5. Find the mistake (Mia). (a) Multiplying scales both parts by the same factor and keeps their comparison the same; adding a fixed amount changes the two parts by different *fractions* of themselves, so the comparison shifts. Here $4 : 6$ simplifies to $2 : 3$, but Mia's $6 : 8$ simplifies to $3 : 4$ — a different ratio. [Manual review: credit any explanation contrasting scaling with adding.]

(b) The water went from 4 cups to 6 cups, a factor of $\frac{6}{4} = 1.5$. Multiply the powder by the same factor: $6 \times 1.5 = 9$ scoops. Check: $6 : 9$ simplifies to $2 : 3$, matching $4 : 6$. Mia's 8 scoops would give $6 : 8 = 3 : 4$, which is not equivalent.

6. Comparing two cars. First car: $150 \div 5 = 30$ miles per gallon. Second car: $210 \div 7 = 30$ miles per gallon. The unit rates are the same, so the two ratios are equivalent: $\boxed{=}$. Unit rate is the fastest equivalence test — both ratios collapse to the same “per one” value, here 30 miles for each single gallon.

7. Find the mistake (Sam). (a) The unit named last in “pages per minute” is the one you divide BY, so pages go on top and minutes on the bottom. [Manual review: credit “the quantity after *per* is the divisor”.]

(b) Correct rate: $\frac{24 \text{ pages}}{3 \text{ minutes}} = \boxed{8}$ pages per minute. Sam computed $\frac{3}{24} = 0.125$, which is the reversed rate — it answers “minutes per page”. A quick sanity check settles it: the printer clearly prints more than one page each minute, so a rate below 1 must be upside down.

8. Challenge: the row that does not belong. (a) Compare the last column with the first: $\frac{8}{22} = \frac{4}{11} \approx 0.36$ while $\frac{2}{5} = 0.4$. So $\boxed{<}$, and the column is not equivalent. The other columns all reduce to $\frac{2}{5}$, so the entry 22 is the wrong one.

(b) Blue paint went from 2 to 8, a factor of 4, so white paint must also be multiplied by 4: $5 \times 4 = 20$. Solving $\frac{2}{5} = \frac{8}{x}$ gives $2x = 40$ and $\boxed{x = 20}$. The corrected column is $8 : 20$, which reduces to $2 : 5$ like every other column.